

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Chapter 10 — Evolution

ANSWER KEY (Version 1)

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Q1. Evolution is defined as the gradual change in a population's traits in response to:

B — Environmental conditions 1 mark

Evolution is the gradual alteration of a population's or species' traits over time in response to the **environmental conditions**. Evolutionary changes can always be passed on to next generations. Food supply, geographic barriers, and reproductive rate are contributing factors but are not the definition.

Q2. Which is NOT a listed source of variation: **C — Photosynthesis** 1 mark

The textbook lists five sources: **genetic recombination** (crossing over), independent assortment of chromosomes, fertilization, **mutation in gamete-forming cells**, and **gene flow**. Photosynthesis is a metabolic process and is not a source of variation.

Q3. Darwin's voyage was on the ship: **B — HMS Beagle** 1 mark

After a **five-year voyage** around the world on the **HMS Beagle**, Charles Darwin put forth the Theory of Natural Selection in 1838. In 1859 he published *On the Origin of Species by Natural Selection*. Victory, Endeavour, and Discovery were other famous ships unrelated to Darwin.

Q4. An inherited trait that increases survival chances is called: **C — Adaptation** 1 mark

An **adaptation** is an inherited trait (or group of traits) that increases an organism's chances of survival in specific environmental conditions. Variation refers to differences in traits; mutation is a gene change; speciation is the formation of new species.

Q5. What most accurately measures an organism's fitness: **B — How many fertile offspring it produces** 1 mark

In the context of **natural selection**, fitness is measured by reproductive success: an organism that produces more **fertile offspring** passes more of its genes to the next generation. Withstanding extremes or eating well are useful but secondary to actual reproduction.

Q6. Peppered moths before 1848 were mostly: **B — Light colour with dark patches** 1 mark

All reported variants of the peppered moth up until 1848 had **light colours with dark patches**. A dark-coloured moth was first discovered in 1848. By 1895 dark-coloured moths made up 95% of the population due to Industrial Melanism.

Q7. Limb bones of humans, dogs, birds, whales with same structure are: **C — Homologous structures** 1 mark

Homologous structures share the same fundamental shape due to origin from a common ancestor, though their shapes and sizes have evolved for different functions. Vestigial structures are inactive remnants; analogous structures have similar function but different origin (not in this textbook).

Q8. A mutation is: **C — A change in gene** 1 mark

A **mutation** is a change in the **gene**. Mutations in gamete-forming germ cells can be inherited and serve as a source of variation. A change in appearance may be the result of a mutation but is not the definition; a change in behaviour is not a mutation.

Q9. Neli-Ravi buffalo were produced by: **C — Selective breeding** 1 mark

Selective breeding (artificial selection) is the practice of breeding domesticated animals to produce offspring with desired genetic features. Cows/buffalo producing more milk are chosen as parents for the next generation, creating breeds that outperform the original.

Q10. Speciation is the evolutionary process by which: **B — A new species is formed** 1 mark

Speciation is defined as the evolutionary process by which new biological **species** arise. A breed is produced by artificial selection; a race and a population are not new biological species by definition.

Q11. Vestigial organs are: **B — Inactive; remains from a previous common ancestor** 1 mark

Vestigial organs are inactive, non-functional structures that are likely just the remains from a previous common ancestor. Examples include wings on flightless birds, leaves on some cacti, and the appendix in humans. They are not newly evolved, not found only in plants, and are distinct from homologous structures.

Q12. Natural selection operates to produce changes in: **B – Populations** **1 mark**

Natural selection causes **population-wide** changes in gene frequencies over time. It does not act on individuals (individuals don't evolve; they live or die). It operates at the level of the whole population, shifting how common or rare certain traits are over generations.

SECTION B – Short Questions (11 × 3 = 33 Marks)

2(i) Define evolution. How old is the Earth and how did life develop? **3 marks**

Evolution is the gradual alteration of a population's or species' traits over time in response to environmental conditions. **1 mark**

According to scientists, the Earth is more than **four billion years old**. **1 mark**

New life forms developed from unicellular organisms, and all current living things developed gradually from earlier ones. Evolutionary changes can always be passed on to next generations. **1 mark**

OR

Define variation. List any three sources of variation from meiosis and related processes. **3 marks**

Variation refers to the differences in traits displayed by members of the same species (e.g., domestic cats differ in size, coat colour, eye colour, hair length). **1 mark**

Three sources of variation: **1 mark each (any 3)**

- **Genetic recombination** from crossing over during gamete development.
- Different chromosome combinations formed by **independent assortment** in gametes.
- The process of **fertilization** (one of countless sperms fuses with an egg).
- A **mutation** in the gamete-forming germ cells.
- **Gene flow** (transfer of genes from one population of a species to another).

2(ii) What is adaptation? How does it differ from variation? Give one example. **3 marks**

An **adaptation** is an inherited trait or group of traits that increases an organism's chances of survival in specific environmental conditions. **1 mark**

Variation refers to differences in traits among individuals of the same species; adaptation is specifically those variations that are **beneficial for survival and reproduction** in a given environment. **1 mark**

Example: the light colour of pre-industrial peppered moths blended with lichen-covered tree bark, giving excellent concealment from predator birds. **1 mark**

OR

How does variation promote evolution? **3 marks**

Some individuals in a population may have an advantage over others as a result of population variations. **1 mark**

These individuals will be able to have more progeny. The population will gradually see more instances of that specific variant. **1 mark**

With these variations, individuals will have a better chance of surviving and procreating, which is evolution. Only the healthiest, fittest, and strongest will survive and reproduce when resources are scarce. **1 mark**

2(iii) State the four bases of Darwin's Theory of Natural Selection. 3 marks

Darwin's Theory of Natural Selection is based on four points **1 mark for correctly naming all four** :

1. **Overproduction:** All living creatures are very fertile so that their populations could grow quickly to produce large numbers of offspring.
2. **Competition:** Population sizes stay roughly constant despite overproduction; food and other resources are scarce, causing competition.
3. **Variations:** No two organisms are the same; a large portion of this diversity is heritable, inherited through genes.
4. **Natural selection:** Only organisms with desirable variations will live till reproductive age and reproduce. Only fittest organisms survive. **2 marks for description**

OR

Darwin's observations at the Galapagos Islands. 3 marks

Darwin visited the **Galapagos Islands** and found a number of **finch species** that varied from island to island.

1 mark

He noticed that the form of the birds' beaks varied depending on the **kind of food they ate** (seeds, insects, cactus, etc.). **1 mark**

Darwin believed the processes of **speciation and adaptation were linked**: accumulation of adaptations in different environments, isolated by geographic barriers, would form a new species from an ancestral form.

1 mark

2(iv) Publication of "On the Origin of Species" and the year the theory was put forth. 3 marks

After a five-year voyage around the world on the HMS Beagle, Charles Darwin put forth the **Theory of Natural Selection in 1838**. **1 mark**

In **1859**, he released the book titled *On the Origin of Species by Natural Selection*, recording his observations.

1 mark

The book described the theory that organisms with favourable traits survive, reproduce more, and pass those traits to offspring, gradually changing the population over generations. **1 mark**

OR

Describe overproduction and competition in natural selection. 3 marks

Overproduction: All living creatures are very fertile so that their populations could grow quickly to produce a large number of offspring. **1.5 marks**

Competition: Population sizes do not rise to very high levels even after overproduction because food supplies and other resources are scarce. As a result, there is competition for resources and food. **1.5 marks**

2(v) Industrial Melanism: moth population before and after the Industrial Revolution. 3 marks

Industrial Melanism is the name given to the phenomenon where dark-coloured moths became dominant in England during the industrial era of pollution. **1 mark**

Before the Industrial Revolution (before 1848): All reported variants of the peppered moth had light colours with dark patches; the dark mutation was first discovered in 1848. **1 mark**

After the Industrial Revolution (by 1895): Dark-coloured moths made up 95% of the peppered moth population in England, as sulphur dioxide pollution from coal burning drove out the lichens and exposed the darker bark of trees. **1 mark**

OR

Why light-coloured moths became easy prey after the Industrial Revolution. 3 marks

Light-coloured **lichens** growing on tree trunks had blended with the colour of the light-coloured moth, giving it excellent concealment from predator birds. **1 mark**

As the industrial revolution progressed, sulphur dioxide pollution from coal burning **drove out lichens**, exposing the darker bark. Light-coloured moths were now visible on dark-coloured trunks and were easily seen and eaten by the birds. **1 mark**

The predator birds served as **snoopers** (agents of natural selection): they ate more light moths on dark trunks, so the dark colour became the adaptive advantage in polluted environments. **1 mark**

2(vi) Three conclusions from the peppered moth case study. 3 marks

The analysis of the peppered moth case study shows three conclusions **1 mark each**:

1. The variation (change in bark colour) is an illustration of **variation subject to selection** by the environment in a changing environment.
2. It demonstrates the **truth of evolution**: populations do change over time in response to environmental pressure.
3. It points out that **evolution and natural selection operate on the basis of genetic change** within a population.

OR

After the 1950s pollution was reduced; what happened to the moths? 3 marks

After the 1950s, England's industrial pollution was reduced. **1 mark**

As pollution decreased, lichens returned to tree trunks, making the bark lighter again. The light-coloured moths now had better **camouflage** than the dark moths. **1 mark**

As a result, the **number of light-coloured moths increased** again. This shows that natural selection is reversible when environmental conditions change, confirming the mechanism of selection by predators. **1 mark**

2(vii) Fossil record and one textbook example of evolution shown by fossils. 3 marks

The **fossil record** is the collection of all fossils from around the world, classified and aged by scientists. It shows the evolution of organisms across time and tells the tale of the past over millions of years. **1 mark**

Fossils also show that ancient species were distinct from modern ones, providing strong evidence that ancient species were distinct from modern ones. **1 mark**

The textbook mentions that researchers have found comprehensive records demonstrating how **humans and horses** have evolved over time. **1 mark**

OR

What are vestigial organs? Give two examples. 3 marks

Vestigial organs are inactive, non-functional structures that organisms possess and which are likely just the remains from a previous common ancestor. These structures have lost their use over time. **1 mark**

Example 1: **Wings on flightless birds** (e.g., ostriches) are vestigial; the ancestor could fly but modern birds of this type cannot. **1 mark**

Example 2: The **appendix in humans** is a vestigial organ with no known current digestive function, believed to be a remnant from ancestors with a different digestive system. (Also acceptable: leaves on some cacti.) **1 mark**

2(viii) Homologous structures with four animals. 3 marks

Homologous structures are similar parts in different animals that share the same fundamental shape due to their origin from a common ancestor. The shapes and sizes have evolved over time in different animals but the general structure remains the same. **1 mark**

Example: the bones in the limbs of a **human, dog, bird, and whale** all share the same general structure (humerus, radius, ulna, and hand/foot bones). **1 mark**

This similarity in structure is evidence of **common ancestry** and supports the theory of evolution through comparative anatomy. **1 mark**

OR

Comparative anatomy as evidence of evolution. 3 marks

Comparative anatomy involves the study of structural similarities in creatures of different species. The existence of structures with the same fundamental shape is a type of evidence supporting evolution. **1 mark**

Due to their origin in the appendages of a **common ancestor**, the bones in the limbs of different mammals share the same general structure, even though their shapes and sizes have evolved differently. **1 mark**

Sharing the same fundamental bone structure indicates descent from a common ancestor, meaning these diverse animals evolved from one ancestral form, which is the definition of evolution. **1 mark**

2(ix) What is artificial selection? Name its two techniques. 3 marks

Artificial selection is the practice of breeding domesticated and genetically enhanced plants and animals to produce offspring with desired genetic features. **1 mark**

Artificial selection is also called **selective breeding**. **1 mark**

The two techniques used in artificial selection are: **selective breeding** and **hybridization**. **1 mark**

OR

Two examples of improving plants by selection; effect on yield. 3 marks

Farmers select plant varieties that cause plants to grow **larger, more quickly, be drought-resistant**, or have higher nutritional value. **1 mark**

Example 1: **Corn (maize)** was bred from a small wild plant (teosinte) through generations of selection; modern corn is much larger. **1 mark**

Over generations of repeating the selection process (choosing seeds only from individuals with the most desirable traits), the **yield of the crop increases** progressively. **1 mark**

2(x) Define speciation and the role of geographical isolation. 3 marks

Speciation is the evolutionary process by which new biological species arise. **1 mark**

When a species gets split into two **geographical regions**, the process of speciation starts. Each group adapts to its own environmental conditions independently. **1 mark**

After a while, individuals from one group can no longer reproduce with the other group (reproductive isolation). Ultimately, two new species have evolved from one ancestral species. **1 mark**

OR

Reproductive isolation leading to two distinct species. 3 marks

Reproductive isolation occurs when two groups that originated from the same ancestor can no longer interbreed successfully. **1 mark**

Once a species is separated by geographical barriers, each group accumulates different adaptations over many generations. Their gene pools diverge. **1 mark**

Eventually the two groups differ so much genetically that they can no longer reproduce with each other, meaning they have become **two distinct species**. Ultimately, two new species have evolved from one. **1 mark**

2(xi) Three differences between natural selection and artificial selection. 3 marks

Natural Selection	Artificial Selection
Selection occurs when natural environmental conditions change.	It is selected by man. 1 mark
Varieties are produced by mutations.	Varieties are produced by selective breeding. 1 mark
It is a slow process and takes thousands to millions of years.	It is a fast process; results are immediate (in the next generation). 1 mark

OR

How does selective breeding improve animals? One example. 3 marks

Animals are subjected to selection by choosing individuals with the most desirable traits as parents for the next generation. Over many generations, offspring that carry and express those traits become the dominant form of the breed. **1 mark**

Example: **Cows** that generate a lot of milk and high-quality meat are chosen for selective breeding. They serve as parents for the following generation. **1 mark**

In this manner, a breed of cow is created that can produce **more milk** than the original breed, demonstrating improvement by artificial selection. **1 mark**

3(i) Darwin's Theory of Natural Selection in detail (four bases). 5 marks

Natural selection is the process through which organisms who possess traits that are helpful for reproduction in a certain environment produce more children in the following generation, gradually expanding the population's **genetic diversity**. 1 mark

The theory is based on four interconnected points:

1. **Overproduction:** All living creatures are very fertile so their populations could grow quickly to produce large numbers of offspring. This means not all can survive. 1 mark
2. **Competition:** Population sizes do not rise to very high levels even after overproduction because food supplies and other resources are scarce. This creates a struggle for existence. 1 mark
3. **Variations:** No two organisms are the same; a large portion of this diversity is heritable, inherited through the genes of offspring. Those with favourable variations have a survival advantage. 1 mark
4. **Natural selection:** Only those organisms with desirable variations will live until reproductive age. The population will have more advantageous genetic variations and fewer harmful ones after many generations. Natural selection is essentially an elimination process; only the fittest organisms survive. 1 mark

OR

Darwin's observations on the HMS Beagle voyage and the Galapagos finches. 5 marks

After a **five-year voyage** around the world on the ship HMS Beagle, Charles Darwin put forth the Theory of Natural Selection in 1838. 1 mark

Darwin was able to create his theory when he visited the **Galapagos Islands**. He found a number of finch species that varied from island to island. 1 mark

He noticed that the form of the birds' **beaks varied depending on the kind of food they ate** (woodpecker finch used tools, seed-eaters had thick beaks, insect-eaters had slender beaks, etc.). 1 mark

Darwin believed that **speciation and adaptation were linked**: the overall accumulation of adaptations in different environments, isolated from the original environment by geographic barriers, would lead to the formation of a new species from an ancestral form. 1 mark

Darwin identified three basic patterns in biological diversity: (1) **global**, (2) **local**, and (3) **variation over time in species**. The two groups could differ sufficiently over many generations to be classified as two different species.

1 mark

3(ii) Peppered moth case study: colour shift during and after the Industrial Revolution. 5 marks

Pre-industrial phase (before 1848): All peppered moths (*Biston betularia*) had light colours with dark patches. The moths rest on tree trunks throughout the day. Light-coloured lichens growing on the trunks blended with the colour of the light-coloured moth, giving excellent concealment from predator birds. Dark moths were rare (from a mutation). **1 mark**

During the Industrial Revolution (1848 onward): Sulphur dioxide pollution from coal burning drove out lichens from trees in industrial locations, exposing the darker bark. Light-coloured moths resting on dark-coloured trunks were now easily visible. Predator birds could locate and consume the light-coloured moths easily. **1 mark**

Selection mechanism: Dark colour was the adaptive variable: dark moths resting on dark trunks were hidden from the birds. Predator birds served as **snoopers**, picking off the visible light moths. By 1895, dark-coloured moths made up **95% of the population**. This phenomenon is called **Industrial Melanism**. **2 marks**

Post-1950s reversal: After England's industrial pollution was reduced, lichens returned to tree trunks, the bark lightened, and the light-coloured moths regained their camouflage advantage. The number of light-coloured moths **increased again**, confirming that natural selection reverses when the environment changes back. **1 mark**

OR

Three types of evidence for evolution with examples. 5 marks

1. Palaeontology (Fossil Record): Fossils show the evolution of organisms across time. Scientists classify and age fossils from around the world; the resulting fossil record tells the tale of the past over millions of years. Example: comprehensive fossil records demonstrate how **humans and horses** have evolved. **2 marks**

2. Comparative Anatomy:

- **Homologous structures:** Structures in creatures with the same fundamental shape due to origin from a common ancestor, e.g., limb bones of human, dog, bird, and whale all share the same general bone arrangement. **1 mark**
- **Vestigial structures:** Inactive, non-functional remains from a previous common ancestor, e.g., wings on flightless birds, the appendix in humans. **1 mark**

3. Selective Breeding: The practice of breeding domesticated plants and animals to produce offspring with desired genetic features demonstrates that traits are heritable and can be directed. Example: **modern corn** was developed from teosinte through generations of selection by farmers. **1 mark**

3(iii) Natural selection vs artificial selection: five key differences and one example each. 5 marks

Natural Selection	Artificial Selection
Selection occurs when natural environmental conditions change.	It is selected by man.
Varieties are produced by mutations.	Varieties are produced by selective breeding.
It operates in natural population.	It operates in domesticated population.
It is a slow process and takes thousands to millions of years.	It is a fast process; results are immediate (in the next generation).
It is responsible for the great biological diversity.	It is responsible for new varieties different from the starting generation.

1 mark per row, any 5 correct differences = 3 marks; 1 mark per valid example = 2 marks

Example of natural selection: The Industrial Melanism of the peppered moth, where dark moths became dominant in polluted environments without human intervention.

Example of artificial selection: Dog breeds (e.g., Labrador, Dachshund, Poodle) were all produced from wolf ancestry by humans selecting for specific traits over generations.

OR

Artificial selection: four-step crop process and one animal example. 5 marks

Artificial selection is the practice of breeding domesticated and genetically enhanced plants and animals to produce offspring with desired genetic features. Its two techniques are selective breeding and hybridization.

1 mark

The four-step process of increasing crop yield by artificial selection:

1. **Natural variation** occurs in the wild population of the crop. **1 mark**
2. Seeds for the next generation are **chosen only from individuals with the most desirable traits** (e.g., largest grain, drought resistance).
3. This process is **repeated for several generations**, each time selecting only the best individuals.
4. Over time, the **yield of the crop increases**. **1 mark** (Steps 2-4 together = 2 marks)

Animal example: Cows that generate a lot of milk and high-quality meat are chosen for selective breeding. They serve as parents for the following generation. In this manner, a breed of cow is created that can produce **more milk** than the original breed. **1 mark**

3(iv) Speciation: geographical isolation, reproductive isolation, and Darwin's three patterns. 5 marks

Speciation is the evolutionary process by which new biological species arise. Two groups that originate from the same ancestor can become very different over time if they live in different places. **1 mark**

Geographical isolation: When a species gets split into two geographical regions, the process of speciation starts. Each group adapts to its own environmental conditions independently. **1 mark**

Reproductive isolation: After a while, individuals from one group can no longer reproduce with the other group. The two populations' gene pools have diverged so much that interbreeding fails. **1 mark**

Result: Ultimately, two new species have evolved from one ancestral species. The two groups differ sufficiently over many generations to be classified as two different species. **1 mark**

Darwin's three patterns of biological diversity: (1) **global** variation (differences between species across the whole world), (2) **local** variation (differences between species in one area), and (3) **variation over time** (species change their form across geological time). **1 mark**

OR

Variation and its role in evolution: definition, five sources, adaptation, and heritable change. 5 marks

Variation is the differences in traits displayed by members of the same species. For instance, domestic cats differ in size, coat colour, eye colour, and hair length. **1 mark**

Five sources of variation (from meiosis and related processes): **1 mark**

1. Genetic recombination from crossing over during gamete development.
2. Different chromosome combinations by independent assortment.
3. Fertilization (one of countless sperms fuses with an egg).
4. Mutation in the gamete-forming germ cells.
5. Gene flow (transfer of genes between populations).

Adaptation: An inherited trait or group of traits that increases an organism's chances of survival in specific environmental conditions is called an adaptation. Variations are beneficial for organisms as they help them be more advanced and immune. **1 mark**

Heritable variation drives evolutionary change: A large portion of variation is heritable, inherited from parents through the genes of offspring. Individuals with advantageous variations survive, reproduce more, and pass those traits on. Over many generations, the population changes, which is evolution. **2 marks**